

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (IT/CE)

Nov 2012

IT402: Wireless communication and Mobile Computing

Date: 05.12.2012, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 Write the answer of following Questions. [11]**
- (a) What is delay spread in multipath propagation? [1]
 - (b) What are the frequencies allocated to GSM? [1]
 - (c) Why can waves with a very low frequency follow the earth's surface? Why are they not used for data transmission in computer networks? [2]
 - (d) Draw protocol stack for GSM. [2]
 - (e) Why the baseband signals cannot be directly transmitted in a wireless system? [2]
 - (f) Explain terms: Shadowing, Reflection and Refraction of signals [3]
- Q - 2 Write the answer of following Questions. [12]**
- (a) What are the differences and similarities between CDMA and 3G? [5]
 - (b) What are the advantages and disadvantages of cellular system with small cell size? [7]
- OR**
- (b) Draw and explain architecture of GSM. Explain entities and responsibilities of GSM architecture? [7]
- Q - 3 Write the answer of following Questions. (Attempt any TWO) [12]**
- (a) What is Hidden and exposed Terminal? What are the possible solutions?
 - (b) How do FHSS and DSSS work?
 - (c) Write a short note on Radiation patterns of directed and sectorized antennas.

SECTION - II

- Q - 4 Write the answer of following Questions. [11]**
- (a) With respect to Mobile IP, What is difference between agent advertisement and agent solicitation? [2]
 - (b) TCP is a connection oriented protocol. It was developed while considering wire line network. Give your suggestion for following parameters to be adapted into Wireless network. [2]
 - i) Window Size
 - ii) MTU size
 - (c) Explain Terms: BSS, ESS [2]
 - (d) What is Transaction Oriented TCP? What is application of T-TCP? [2]
 - (e) What is role played by Foreign Agent with respect to Mobil IP? [3]
- Q - 5 Write the answer of following Questions. [12]**
- (a) Draw and explain fields of 802.11 MAC packet structure. [5]
 - (b) What are the several steps suggested in Indirect TCP when mobile station moves to other network. Write down the advantages and disadvantages of Indirect TCP. [7]
- OR**
- (b) Write down Bluetooth Protocol Stack. Explain functionalities of each layer. [7]

Q - 6 Write the answer of following Questions. (Attempt any TWO)

[12]

- (a) What are the advantages and disadvantages of Wireless LAN?
- (b) Write a short note on: HYPERLAN
- (c) What are the differences between wired network and wireless network with respect to routing?